

ZHONGHAO LIN

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EDUCATION

Beijing Institute of Technology (BIT), Beijing, China

Sept. 2024 – Jun. 2027 (Expected)

M.ENG. in CONTROL ENGINEERING

GPA: 92.4 / 100, 3.9 / 4.0

Selected Graduate Coursework: Linear Algebra in Automatic Control (97 / 100), Optimization Theory and Methods (96 / 100), Multi-Source Information Filtering and Fusion (95 / 100), Multi-Agent Coordination and Control (94 / 100), Linear System Theory (92 / 100)

Advisor: Prof. Xianlin Zeng

Research Topics: Time-varying optimization, online optimization, distributed optimization, and their applications to localization, tracking, and networked systems.

Beijing Institute of Technology (BIT), Beijing, China

Sept. 2020 - Jun. 2024

B.ENG. in AUTOMATION

GPA: 91.3 / 100, 3.8 / 4.0 (ranked top 3%)

Selected Coursework: Mathematical Analysis (99 / 100), Linear Algebra (99 / 100), Complex Functions (99 / 100), Operations Research (99 / 100), Probability Theory (98 / 100), Intelligent Control (97 / 100), Automatic Control Theory (96 / 100), Information Theory (95 / 100), Signal Analysis and Processing (94 / 100).

PUBLICATIONS AND MANUSCRIPTS

1. **Z. Lin**, J. Hou, and X. Zeng, "Optimal Prediction-Correction Algorithm Using Sparse Linear Extrapolation for Time-Varying Optimization," *IEEE Transactions on Signal Processing*, vol. 74, pp. 2564-2578, 2026.
2. **Z. Lin**, X. Zeng, J. Hou, J. Sun, and J. Chen, "Primal-Dual Prediction-Correction Method with Tunable Memory for Linearly Constrained Time-Varying Convex Optimization," *Journal of Systems Science and Complexity*, vol. 39, no. 2, pp. 483-510, 2026.
3. **Z. Lin**, J. Hou, and X. Zeng, "Hessian-Inverse-Free Prediction-Correction Method for Time-Varying Convex Optimization," accepted in *IEEE Conference on Decision and Control (CDC 2026)*.
4. S. Wang, **Z. Lin**, X. Zeng, "Distributed Optimization with Anytime Coupling Constraints and Asymptotic Consensus," accepted in *International Conference on Automation, Control and Robotics (ICACR 2026)*.
5. "Linear-Extrapolation Prediction-Correction for TOA-Based Target Tracking," working paper.

RESEARCH EXPERIENCE

Prediction-Correction Methods for Time-Varying Optimization (Master's thesis research)

Advisor: Prof. Xianlin Zeng, BIT, China

Sept. 2024 – Present

- Proposed a sparse linear extrapolation prediction-correction (SE-PC) method for strongly convex time-varying optimization, where predictor coefficients are designed via ℓ_1 -norm minimization under tractable constraints. The method attains state-of-the-art tracking accuracy while substantially reducing the number of gradient-descent correction steps required for convergence. This work has been published in *IEEE Transactions on Signal Processing*.
- Developed a primal-dual prediction-correction (PD-PC) method with tunable memory for linearly constrained time-varying convex optimization. The method combines tunable-memory extrapolation with primal-dual gradient correction and achieves second-order asymptotic tracking accuracy under time-varying linear constraints. This work has been published in *Journal of Systems Science and Complexity*.
- Proposed a Hessian-inverse-free prediction-correction flow for continuous-time time-varying convex optimization. The method introduces an auxiliary matrix dynamics to generate an online surrogate of the Hessian inverse, thereby avoiding explicit matrix inversion. This work is currently accepted for presentation at the *IEEE Conference on Decision and Control (CDC 2026)*.
- Extended the linear-extrapolation prediction-correction approach from the previous TSP work to TOA-based target tracking. Established that both extrapolated predictions and corrected iterates remain within

the local strong-convexity region of the current TOA least-squares objective. Derived an explicit tracking bound showing Q -linear convergence to a noise-dependent neighborhood, which vanishes in the noiseless case. This manuscript is being prepared for future submission to *IEEE Signal Processing Letters*.

Distributed Optimization-Based Cooperative Velocity Control (Bachelor's thesis research)

Advisor: Prof. Xianlin Zeng, BIT, China

Sept. 2023 – Jun. 2024

- Formulated the cooperative velocity control problem of multiple autonomous vehicles as a consensus-based distributed optimization problem, where fuel consumption was modeled as the objective function and safety and communication requirements were incorporated as time-varying constraints.
- Developed a distributed velocity control algorithm by combining projected gradient methods, control barrier functions, and sliding-mode control, and validated its convergence and robustness through multi-vehicle simulations.

SELECTED HONORS AND AWARDS

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| 1. Special-Class Graduate Scholarship (ranked top 5%), BIT. | 2024 |
| 2. Outstanding Graduate , BIT. | 2024 |
| 3. Samsung Scholarship (ranked 3rd among 200+ students), China Samsung. | 2023 |
| 4. First Prize , Chinese Mathematics Competitions for College Students, awarded twice. | 2021, 2022 |
| 5. First-Class Undergraduate Scholarship (ranked top 5%), BIT, awarded four times. | 2021–2024 |
| 6. First Prize , BIT Freshman Mathematics Competition. | 2020 |

ACADEMIC ACTIVITIES & SERVICE

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| 1. Presenter (only master's student presenter), <i>PhD Workshop on Frontiers in Optimization Algorithms, Software, and Applications</i> . | 2026 |
| 2. Reviewer, <i>IEEE International Conference on Control and Automation (ICCA 2026)</i> . | 2026 |
| 3. Reviewer, <i>Chinese Control Conference (CCC 2026)</i> . | 2026 |

SKILLS

- **Languages:** Mandarin Chinese (native), English (CET-6: 580 / 710, IELTS: 7.0).
- **Programming and Tools:** MATLAB, python, $\text{\LaTeX}$.